

HW #5

5.7

A)

AB	00	01	11	10
00	1	1		1
01				
11	1	1		
10	1		1	

$$\begin{aligned} \overline{A}BCD + A\overline{B}CD &= \overline{A}BC \\ \overline{A}BCD + A\overline{B}CD &= \overline{A}BC \\ \overline{A}BCD + A\overline{B}CD &= \overline{A}BC \\ \overline{A}BCD & \\ \overline{A}BCD & \end{aligned}$$

$$\overline{A}BC + \overline{A}BC + \overline{A}BC + \overline{A}BC + \overline{A}BC$$

B)

AB	00	01	11	10
00	1	1		
01	1			
11	1		1	
10	1		1	

$$\begin{aligned} \overline{A}BCD + A\overline{B}CD &= \overline{A}B \\ \overline{A}BCD + A\overline{B}CD &= \overline{A}B \\ \overline{A}BCD + A\overline{B}CD &= \overline{A}B \\ \overline{A}BCD & \\ \overline{A}BCD & \end{aligned} \rightarrow \overline{A}B + \overline{A}B + \overline{A}B + \overline{A}B + \overline{A}B$$

1011
1010

	00	01	11	10
00	1	0	1	1
01	0	1	1	0
11	0	1	0	1
10	0	1	1	1

$$\begin{aligned} \overline{A}BCD + A\overline{B}CD &= BCD \\ \overline{A}BC\overline{D} + A\overline{B}C\overline{D} &= \overline{A}C\overline{D} \\ \overline{A}BCD + \overline{A}BC\overline{D} &= \overline{A}BC \\ \overline{A}BCD + \overline{A}B\overline{C}D &= \overline{A}BD \\ \overline{A}BC\overline{D} + \overline{A}B\overline{C}\overline{D} &= \overline{A}B\overline{D} \end{aligned}$$

$$\overline{B}CD + \overline{A}C\overline{D} + \overline{A}BC\overline{D} + \overline{A}BD + \overline{A}B\overline{D} + \overline{A}BC + \overline{A}B\overline{C}$$

	00	01	11	10
00	0	0	X	0
01	X	1	1	X
11	1	1	X	1
10	0	0	1	1

0101 0101 0111 1110 1011
1101 0111 0011 1010 1010

$$P = (D + Ae)$$

5.12

	00	01	11	10
00	0	1	0	1
01	0	1	0	0
11	1	1	1	1
10	1	1	0	1

$$F = \prod M(0, 1, 9, 12, 13, 14)$$

$$F = (\overline{A}BCD)(\overline{A}BC\overline{D})(\overline{A}B\overline{C}D)(\overline{A}B\overline{C}\overline{D})(\overline{A}B\overline{C}D)(\overline{A}B\overline{C}\overline{D})$$

$$b) F = \overline{A}BC + \overline{A}B\overline{C} + \overline{A}B\overline{D} + \overline{A}B\overline{C}\overline{D}$$

$$c) F = (A+B+C)(A+B+C)(\overline{A}+\overline{B}+\overline{D})(\overline{A}+\overline{B}+\overline{C}+\overline{D})$$

5.14

A \ BC	00	01	11	10
00	0	0	0	0
01	1	1	0	0
11	0	1	1	0
10	1	0	0	0

$$\begin{aligned} \overline{A}BC + A\overline{B}C &= \overline{B}C \\ A\overline{B}C + A\overline{B}\overline{C} &= A\overline{B} \\ A\overline{B}C + A\overline{B}\overline{C} &= A\overline{B} \end{aligned} \Rightarrow \overline{B}C + A\overline{B}C + \overline{A}B\overline{C}$$

5.15

A \ ST	00	01	11	10
00	1	1	0	0
01	1	1	0	0
11	0	0	1	1
10	1	1	0	0

$$\overline{A}\overline{S}T + A\overline{S}\overline{T} + \overline{A}S\overline{T} + A\overline{S}T + A\overline{S}T + \overline{A}S\overline{T}$$

$$B = \overline{R}S + R\overline{S} + 0\overline{R}$$

5.16

A \ PQ	00	01	11	10
00	0	0	1	0
01	1	1	1	0
11	1	1	0	0
10	0	0	0	0

$$f_s = \overline{N}Q + N\overline{P}$$

5.17

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	0
10	1	1	1	0

$$B) F = \overline{A}B + C\overline{D} + ABC$$

$$C) F = (A+B)(A\overline{D})(B+\overline{D})$$

5.19

A	x_1	x_2	z
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

5.20

x_1x_2	00	01	11	10
00	0	0	1	0
01	0	1	0	1
11	1	1	1	0
10	1	1	1	0

$$(\overline{x_1} + x_2)(c_1 + c_2 + x_1)(c_1 + x_1 + x_2)(\overline{c_1} + \overline{c_2} + x_1 + \overline{x_2})$$

$$(c_1 + c_2 + x_1 + \overline{x_2})$$

S.22

c)

	AB	00	01	11	10
00	0	1	0	x	
01	1	x	1	x	
11	0	x	0	0	
10	1	x	1	x	

$$= B\overline{C}\overline{D} + B\overline{C}D + \overline{C}\overline{D}$$

e)

	AB	00	01	11	10
00	0	1	0	x	
01	0	x	0	x	
11	1	x	1	0	
10	0	x	0	x	

$$= \overline{C}D + \overline{B}C$$

f)

	AB	00	01	11	10
00	0	1	x	x	
01	0	x	x	x	
11	0	x	x	x	
10	0	x	x	x	

$$= B$$

5.25

A)

AB \ CD	00	01	11	10
00		1		
01	1	1	1	1
11	1	1	1	1
10				

$$f = \bar{C}D + BD + \bar{A}BD$$

B)

AB \ CD	00	01	11	10
00	0	1	1	0
01	1	0	1	1
11	0	1	1	0
10	1	1	1	1

$$(B + C + D)(B + C + \bar{D})(A + B + C + D)$$

$$\bar{B}C\bar{D} + \bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D}$$

C)

AB \ CD	00	01	11	10
00	1			X
01			1	1
11	X			
10	1	1	1	X

$$\bar{A}BD + \bar{C}D + B\bar{C}\bar{D}$$

D)

AB \ CD	00	01	11	10
00	0		0	
01	X	X	0	0
11	X	0		
10	0	0		

$$(A + B + D)(A + \bar{B} + \bar{C})(\bar{A} + \bar{B} + C)(\bar{A} + C + \bar{D})$$

S.28

AB \ CD	00	01	11	10
00	1	0	0	1
01	1	0	1	1
11	x	1	x	0
10	1	0	0	1

$\Rightarrow \overline{B} + C$

S.34

A)

AB \ CD	00	01	11	10
00	x	1	x	0
01	0	x	0	x
11	x	1	x	1
10	1	x	1	x

B) $C \overline{A}', C, B \overline{C} \in \text{PI}$

C) $\overline{C} + C + B \overline{C} \rightarrow C + B \overline{C} = \overline{B} + C$